РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №16

дисциплина: Администрирование сетевых подсистем

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Группа: НФИбд-02-23

МОСКВА

2025 г.

Цель работы:

Целью данной работы является получение навыков работы с программным средством Fail2ban для обеспечения базовой защиты от атак типа «brute force».

Выполнение работы:

```
  Installing       : fail2ban-server-1.1.0-6.el9.noarch                                  2/5
  Running scriptlet: fail2ban-server-1.1.0-6.el9.noarch                                  2/5
  Installing       : fail2ban-firewalld-1.1.0-6.el9.noarch                               3/5
  Installing       : fail2ban-sendmail-1.1.0-6.el9.noarch                                4/5
  Installing       : fail2ban-1.1.0-6.el9.noarch                                         5/5
  Running scriptlet: fail2ban-selinux-1.1.0-6.el9.noarch                                 5/5
  Running scriptlet: fail2ban-1.1.0-6.el9.noarch                                         5/5
  Verifying        : fail2ban-1.1.0-6.el9.noarch                                         1/5
  Verifying        : fail2ban-firewalld-1.1.0-6.el9.noarch                               2/5
  Verifying        : fail2ban-selinux-1.1.0-6.el9.noarch                                 3/5
  Verifying        : fail2ban-sendmail-1.1.0-6.el9.noarch                                4/5
  Verifying        : fail2ban-server-1.1.0-6.el9.noarch                                  5/5

Installed:
  fail2ban-1.1.0-6.el9.noarch                  fail2ban-firewalld-1.1.0-6.el9.noarch
  fail2ban-selinux-1.1.0-6.el9.noarch          fail2ban-sendmail-1.1.0-6.el9.noarch
  fail2ban-server-1.1.0-6.el9.noarch

Complete!
[root@server ~]# systemctl start fail2ban
[root@server ~]# systemctl enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/
system/fail2ban.service.
[root@server ~]# S
```

```
root@server:~
root@server:~        root@server:~
[vagrant@server ~]$ sudo -i
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:34:42,274 fail2ban.server         [7099]: INFO    --------------------------------
------------------
2026-03-12 20:34:42,274 fail2ban.server         [7099]: INFO    Starting Fail2ban v1.1.0
2026-03-12 20:34:42,275 fail2ban.observer       [7099]: INFO    Observer start...
2026-03-12 20:34:42,295 fail2ban.database       [7099]: INFO    Connected to fail2ban persistent
 database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-03-12 20:34:42,296 fail2ban.database       [7099]: WARNING New database created. Version '4
'
```

customisation.local
/etc/fail2ban/jail.d

```
[DEFAULT]
bantime = 3600

#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
```

```
(gedit:7201): dconf-WARNING **: 20:37:18.502: failed to commit changes to dconf: Failed to execu
te child process “dbus-launch” (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:37:31,227 fail2ban.jail           [7230]: INFO    Initiated 'polling' backend
2026-03-12 20:37:31,229 fail2ban.filter         [7230]: INFO      maxLines: 1
2026-03-12 20:37:31,238 fail2ban.filter         [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,239 fail2ban.filter         [7230]: INFO      findtime: 600
2026-03-12 20:37:31,239 fail2ban.actions        [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,239 fail2ban.filter         [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,245 fail2ban.filtersystemd  [7230]: INFO    [sshd] Jail is in operation now
(process new journal entries)
2026-03-12 20:37:31,246 fail2ban.jail           [7230]: INFO    Jail 'sshd' started
2026-03-12 20:37:31,249 fail2ban.jail           [7230]: INFO    Jail 'selinux-ssh' started
2026-03-12 20:37:31,259 fail2ban.jail           [7230]: INFO    Jail 'sshd-ddos' started
```

```
 database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-03-12 20:37:31,146 fail2ban.jail           [7230]: INFO    Creating new jail 'sshd'
2026-03-12 20:37:31,150 fail2ban.jail           [7230]: INFO    Jail 'sshd' uses systemd {}
2026-03-12 20:37:31,151 fail2ban.jail           [7230]: INFO    Initiated 'systemd' backend
2026-03-12 20:37:31,156 fail2ban.filter         [7230]: INFO      maxLines: 1
2026-03-12 20:37:31,180 fail2ban.filtersystemd  [7230]: INFO    [sshd] Added journal match for:
'_SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session'
2026-03-12 20:37:31,180 fail2ban.filter         [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,180 fail2ban.filter         [7230]: INFO      findtime: 600
2026-03-12 20:37:31,180 fail2ban.actions        [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,180 fail2ban.filter         [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,181 fail2ban.jail           [7230]: INFO    Creating new jail 'selinux-ssh'
2026-03-12 20:37:31,182 fail2ban.jail           [7230]: INFO    Jail 'selinux-ssh' uses poller {
}
2026-03-12 20:37:31,182 fail2ban.jail           [7230]: INFO    Initiated 'polling' backend
2026-03-12 20:37:31,183 fail2ban.datedetector   [7230]: INFO      date pattern `''`: `Epoch`
2026-03-12 20:37:31,184 fail2ban.filter         [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,184 fail2ban.filter         [7230]: INFO      findtime: 600
2026-03-12 20:37:31,184 fail2ban.actions        [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,184 fail2ban.filter         [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,224 fail2ban.filter         [7230]: INFO    Added logfile: '/var/log/audit/a
udit.log' (pos = 0, hash = 0c2be31ffde8214ab616b0e504468ebe00b577f7)
2026-03-12 20:37:31,224 fail2ban.jail           [7230]: INFO    Creating new jail 'sshd-ddos'
2026-03-12 20:37:31,226 fail2ban.jail           [7230]: INFO    Jail 'sshd-ddos' uses poller {}
2026-03-12 20:37:31,227 fail2ban.jail           [7230]: INFO    Initiated 'polling' backend
2026-03-12 20:37:31,229 fail2ban.filter         [7230]: INFO      maxLines: 1
2026-03-12 20:37:31,238 fail2ban.filter         [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,239 fail2ban.filter         [7230]: INFO      findtime: 600
2026-03-12 20:37:31,239 fail2ban.actions        [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,239 fail2ban.filter         [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,245 fail2ban.filtersystemd  [7230]: INFO    [sshd] Jail is in operation now
(process new journal entries)
2026-03-12 20:37:31,246 fail2ban.jail           [7230]: INFO    Jail 'sshd' started
2026-03-12 20:37:31,249 fail2ban.jail           [7230]: INFO    Jail 'selinux-ssh' started
2026-03-12 20:37:31,259 fail2ban.jail           [7230]: INFO    Jail 'sshd-ddos' started
```

customisation.local
/etc/fail2ban/jail.d

```
[DEFAULT]
bantime = 3600

#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true

#
# HTTP servers
#
[apache-auth]
enabled = true
[apache-badbots]
enabled = true
[apache-noscript]
enabled = true
[apache-overflows]
enabled = true
[apache-nohome]
enabled = true
[apache-botsearch]
enabled = true
[apache-fakegooglebot]
enabled = true
[apache-modsecurity]
enabled = true

[apache-shellshock]
enabled = true
```

Saving file "/etc/fail2ban/jail.d/customisation.local"... Plain Text Tab Width: 8 Ln 27, Col 16 INS

```
root@server:~
root@server:~    root@server:~    root@server:~
** (gedit:7296): WARNING **: 20:39:15.499: Set document metadata failed: Setting attribute metad
ata::gedit-position not supported

(gedit:7296): dconf-WARNING **: 20:39:15.517: failed to commit changes to dconf: Failed to execu
te child process “dbus-launch” (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:39:27,417 fail2ban.filter         [7328]: WARNING [apache-botsearch] Please check
a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn]
 [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library
 (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) -
 may result in undefined or erroneous behavior
2026-03-12 20:39:27,431 fail2ban.jail           [7328]: INFO    Jail 'apache-botsearch' started
2026-03-12 20:39:27,440 fail2ban.jail           [7328]: INFO    Jail 'apache-fakegooglebot' star
ted
2026-03-12 20:39:27,448 fail2ban.filter         [7328]: WARNING [apache-modsecurity] Ignoring al
l log entries older than 600s; these are probably messages generated while fail2ban was not runn
ing.
2026-03-12 20:39:27,448 fail2ban.filter         [7328]: WARNING [apache-modsecurity] Please chec
k a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:war
n] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer libra
ry (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070)
 - may result in undefined or erroneous behavior
2026-03-12 20:39:27,449 fail2ban.jail           [7328]: INFO    Jail 'apache-modsecurity' starte
d
2026-03-12 20:39:27,461 fail2ban.filter         [7328]: WARNING [apache-shellshock] Ignoring all
 log entries older than 600s; these are probably messages generated while fail2ban was not runni
ng.
2026-03-12 20:39:27,470 fail2ban.jail           [7328]: INFO    Jail 'apache-shellshock' started
2026-03-12 20:39:27,470 fail2ban.filter         [7328]: WARNING [apache-shellshock] Please check
 a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn
] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer librar
y (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070)
- may result in undefined or erroneous behavior
2026-03-12 20:39:27,508 fail2ban.jail           [7328]: INFO    Jail 'sshd-ddos' started
```

```
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
[root@server ~]# fail2ban-client set sshd maxretry 2
2
[root@server ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
[root@server ~]# fail2ban-client set sshd unbanip <ip-адрес клиента>
-bash: syntax error near unexpected token `newline'
[root@server ~]# fail2ban-client set sshd unbanip 128.45.29.1.1
0
[root@server ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
[root@server ~]#
```

Open ▾ | *customisation.local | /etc/fail2ban/jail.d | Save | ≡ | ×

```
1 [DEFAULT]
2 bantime = 3600
3 ignoreip = 127.0.0.1/8
4 #
5 # SSH servers
6 #
7 [sshd]
8 port = ssh,2022
9 enabled = true
```

```
** (gedit:7438): WARNING **: 20:44:29.558: Set document metadata failed: Setting attribute metad
ata::gedit-position not supported

(gedit:7438): dconf-WARNING **: 20:44:29.584: failed to commit changes to dconf: Failed to execu
te child process “dbus-launch” (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:44:38,221 fail2ban.jail           [7469]: INFO    Jail 'apache-auth' started
2026-03-12 20:44:38,247 fail2ban.jail           [7469]: INFO    Jail 'apache-badbots' started
2026-03-12 20:44:38,251 fail2ban.jail           [7469]: INFO    Jail 'apache-noscript' started
2026-03-12 20:44:38,253 fail2ban.jail           [7469]: INFO    Jail 'apache-overflows' started
2026-03-12 20:44:38,257 fail2ban.jail           [7469]: INFO    Jail 'apache-nohome' started
2026-03-12 20:44:38,294 fail2ban.jail           [7469]: INFO    Jail 'apache-botsearch' started
2026-03-12 20:44:38,317 fail2ban.jail           [7469]: INFO    Jail 'apache-fakegooglebot' star
ted
2026-03-12 20:44:38,360 fail2ban.jail           [7469]: INFO    Jail 'apache-modsecurity' starte
d
2026-03-12 20:44:38,392 fail2ban.jail           [7469]: INFO    Jail 'apache-shellshock' started
2026-03-12 20:44:38,402 fail2ban.jail           [7469]: INFO    Jail 'sshd-ddos' started
```

```
root@server:~                    root@server:/vagrant/provision/server

[vagrant@server ~]$ sudo -i
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d cp -R /etc/
fail2ban/jail.d/customisation.local
mkdir: invalid option -- 'R'
Try 'mkdir --help' for more information.
[root@server server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d
[root@server server]# cp -R /etc/fail2ban/jail.d/customisation.local /vagrant/provision/server/p
rotect/etc/fail2ban/jail.d/
[root@server server]# gedit /vagrant/provision/server

(gedit:7569): dconf-WARNING **: 20:46:46.692: failed to commit changes to dconf: Failed to execu
```

```
Open ▾   protect.sh   Save
/vagrant/provision/server

1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install fail2ban
5  echo "Copy configuration files"
6  cp -R /vagrant/provision/server/protect/etc/* /etc
7  restorecon -vR /etc
8  echo "Start fail2ban service"
9  systemctl enable fail2ban
10 systemctl start fail2ban
```

Вывод:

В ходе выполнения лабораторной работы были получены навыки работы с программным средством Fail2ban для обеспечения базовой защиты от атак типа «brute force».

Ответы на контрольные вопросы:

1. Поясните принцип работы Fail2ban. - **Fail2ban является инструментом для защиты от атак на серверы, основанных на анализе журналов. Он мониторит журналы системы на предмет неудачных попыток входа или других событий, а затем блокирует IP-адреса атакующих с использованием системных средств, таких как iptables. Принцип работы:**

   Мониторинг журналов на предмет определенных событий.

   Обнаружение повторных неудачных попыток входа или других нарушений.

   Динамическое обновление правил брандмауэра для блокировки атакующих IP-адресов.

2. Настройки какого файла более приоритетны: jail.conf или jail.local? - **Настройки файла jail.local имеют более высокий приоритет и перекрывают настройки из jail.conf. Таким образом, если есть конфликтующие настройки, они будут использоваться из jail.local.**

3. Как настроить оповещение администратора при срабатывании Fail2ban?

   - В файле jail.local нужно указать параметр destemail и задать адрес электронной почты, а также параметр action с указанием определенного действия (например, action_mw для отправки почты).

4. Поясните построчно настройки по умолчанию в конфигурационном файле /etc/fail2ban/jail.conf, относящиеся к веб-службе. –

   Пример настроек для веб-службы в файле jail.conf:

   [apache]

   enabled = true

   port = http,https

   filter = apache-auth

   logpath = /var/log/apache*/*error.log

5. Поясните построчно настройки по умолчанию в конфигурационном файле /etc/fail2ban/jail.conf, относящиеся к почтовой службе. –

   Пример настроек для почтовой службы в файле jail.conf:

   [postfix]

   enabled = true

   filter = postfix

   action = iptables-multiport[name=postfix, port="submission,smtps", protocol=tcp]

6. Какие действия может выполнять Fail2ban при обнаружении атакующего IP-адреса? Где можно посмотреть описание действий для последующего использования в настройках Fail2ban? - **Fail2ban может выполнять различные действия, такие как блокировка IP-адреса с использованием брандмауэра, отправка уведомлений, добавление в черные списки и т.д. Описание действий можно найти в конфигурационных файлах в разделе action.**
7. Как получить список действующих правил Fail2ban? - **Используйте команду: fail2ban-client status.**
8. Как получить статистику заблокированных Fail2ban адресов? - **Используйте команду: fail2ban-client status <jail_name>.**
9. Как разблокировать IP-адрес? - **Используйте команду: fail2ban-client set <jail_name> unbanip <ip_address>.**